MSC ARTIFICIAL INTELLIGENCE

DEEP LEARNING AND GENERATIVE AI

CRITICAL APPRAISAL & PROPOSAL

SUBMITTED BY

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Table of Contents

Summary 3

Critical Appraisal 4

Theoretical Foundations 4

Methodology Critique 5

Limitations and Ethical Considerations 6

Proposal for Improvement 6

Technical Extension 7

Justification of the Improvement 8

Expected Outcomes and Hypothesis 8

Part 2 Implementation Plan 9

References 10

Summary

The paper "Generalizable Slum Detection from Satellite Images using Mixture-of-Experts" is based on identifying slums through high-resolution satellite images in several different parts of the world (Lee, et al., 2026). There are many reasons why detecting slums is an important task for urban planning, for allocating humanitarian resources, and for sustainability. Models trained to detect slums from images of one city do not work well when applied to other cities. The reason for this is that there are large variations in how houses are packed together, in roofing material, in the arrangement of buildings within a neighborhood, and in the environment itself. As a result of these differences, the statistical distributions of images are very different from place to place.

To address this problem, the authors proposed GRAM (Generalized Region-Aware Mixture-of-Experts). It is a multi-stage model which incorporates a combination of Mixture-of-Experts (MoE) and Test-Time Adaptation (Wang, et al., 2025). For the first stage, during source training, multiple MoE's learn features specific to various geographic regions while learning a shared backbone. During testing, input is directed toward the best MoE based upon a simple gating function. For the second stage, a separate region classification network determines which of the previously selected MoE's generated the best pseudo-labels. Predictions generated by the previously selected MoE's are then filtered for accuracy prior to performing fine-tuned training without additional supervision.

The experimental results demonstrate that GRAM performs significantly better than standard CNN and segmentation-based approaches when tested on unseen regions (Chen, et al., 2025). Most notably, the results indicate that GRAM was able to perform well even when it had very limited amounts of labeled data from these regions. Thus, GRAM can help improve cross-region performance under real-world deployment conditions.

Critical Appraisal

Theoretical Foundations

The GRAM framework was builtbased on the concepts of Deep Learning, Ensemble Learning, and Domain Adaptation, at its base. The model utilizes an Mixture-of-Experts (MoE) structure, which trains a multitude of specialized "expert" networks to identify unique regional patterns, while utilizing a unified "backbone" to extract the common characteristics of input images. From a theoretical basis, the use of MoEs relates to the bias-variance trade-off and the theoretical basis of ensemble learning. Experts can develop their own unique patterns for identifying regional characteristics, thereby reducing the bias towards a singular representation for a particular region; however, the development of experts may also increase variance if they are not properly regularized.

The model utilizes back-propagation with gradient-based optimization to train the model. Both the shared "backbone" and the expert head(s) utilize simultaneous updates during the training phase (Hanandeh, et al., 2025). A gating mechanism has been developed to allow for soft routing of input images to their respective experts. The gating mechanism allows for the selective allocation of representational capacity by the model, rather than distributing representational capacity uniformly across all input images. Theoretically, the utilization of conditional specialization in the model improves representation learning in cases of domain shifts, as region-specific characteristics of informal settlements (i.e., roof texture, spatial density) can be identified by unique experts.

The model also employs pseudo-label refinement as a form of test-time adaptation. Pseudo-label refinement is a principle employed in semi-supervised and unsupervised domain adaptation, whereby high-confidence predictions are utilized to iteratively modify the model parameters (Zeng, et al., 2024). Although the use of pseudo-label refinement is theoretically sound (the model utilizes prediction consistency as a proxy for reliability), the assumption made by the model that there will be agreement among the experts when they correctly predict an image does not always hold. Agreement among experts may not be indicative of correct predictions, particularly in visually ambiguous areas.

Methodology Critique

While the proposed framework has demonstrated improved cross-regional performance, there are several methodological aspects of the framework that warrant additional criticism.

The fair comparison of baselines is dependent upon the methods utilized to compare them. The authors of the paper employ the proposed framework, GRAM, against standard CNN and segmentation models; however, it is not explicitly stated whether all of the baselines were trained with equivalent hyperparameter tuning or data augmentation techniques. If the MoE framework benefits from additional adaptation techniques that are not employed in the training of the baselines, then the comparison may be influenced by differences in the training protocols of the two architectures.

The construction of the dataset, although diverse in terms of geography, may not be representative of the diversity of slums. The morphology of slums varies not only across continents but also at the neighborhood level within cities (Moonlight, 2025). The authors emphasize multi-continent coverage; however, the authors do not provide information regarding the distribution of labeled samples across regions, nor the consistency of annotation standards across the various datasets. If some cities comprise a large portion of the training distribution, the experts may develop biased representations that favor the majority domain.

The authors provide ablation studies to demonstrate the contributions of individual components (i.e., gating and pseudo-label filtering); however, the extent of these studies appears to be limited. For example, it would be advantageous to investigate the impact of varying the number of experts, and/or employing alternative gating mechanisms. Without comprehensive ablation studies across a variety of architectural configurations, it is difficult to determine the full extent to which the performance of the model is attributable to each of the design choices.

The primary focus of the evaluation of the proposed framework is on metrics related to segmentation accuracy. While segmentation accuracy metrics measure the performance of predictive models, the evaluation of additional analyses (i.e., robustness to severe domain shift, performance under extreme class imbalance) would strengthen empirical validation.

Limitations and Ethical Considerations

From a technical standpoint, the GRAM architecture results in increased computational complexity. Multiple expert heads and adaptation procedures result in increased memory usage and processing time during inference, relative to a single CNN model. Therefore, in environments where resources are scarce, the computational demands associated with GRAM may limit scalability.

Additionally, the utilization of pseudo-label adaptation introduces additional instability to the model (Li, et al., 2023). If the model makes incorrect, yet confident, predictions, the model may further solidify incorrect patterns through self-training, resulting in confirmation bias. This risk is particularly pronounced in heterogeneous urban environments, where visual characteristics of informal settlements may overlap with those of non-slum regions.

Ethically, slum detection systems possess a wide range of implications for society. Misclassification may negatively affect the stigma associated with communities and may improperly influence governmental resource allocation. Furthermore, satellite imagery-based inference possesses implications for privacy and governance, particularly if models are employed without oversight from local communities. Finally, training data may contain regional biases that may marginalize underrepresented geographic contexts.

Proposal for Improvement

While the GRAM framework shows good cross-region performance, it uses a method of pseudo-label refinement at test time for adaptation, which could be unstable. The present version of the model also assumes that when experts predict something with a similar confidence, they are correct. But in areas where there are ambiguities visually, or where the environment is diverse, experts can still agree on false positives; therefore, systematically incorrect decisions will be reinforced.

Furthermore, the proposed model does not provide an explicit quantification of predictive uncertainty, and in high-risk social-impact applications, like detecting slums, quantifying uncertainty is very important. Therefore, if we do not have uncertainty modeling, the system might generate too confident predictions, so stakeholders cannot clearly understand the level of risk when using the system in critical deployments.

Therefore, the proposal intends to resolve the previous issues by introducing an Uncertainty-Aware Mixture-of-Experts (UA-MoE) framework that combines predictive uncertainty estimation in both the gating mechanism and pseudo-label selection process to increase robustness, reliability, and responsible decision-making when operating in a different domain than trained (Luttner, 2023).

Technical Extension

The technical extension to this project has two key parts:

1. **Monte Carlo Dropout-based Uncertainty Estimation**

Each expert head will incorporate Monte Carlo (MC) Dropout during inference. For each input image, instead of producing one deterministic prediction that is the same every time, multiple stochastic forward passes are performed to generate a set of possible predictions (Son & Seok, 2026). Let $p^{(t)}$ denote the prediction of expert i during stochastic pass t. The predictive mean and variance are computed as:

$$\boldsymbol{\mu_i} = \frac{1}{T}\sum_{\boldsymbol{t=1}}^{T} p_i^{(t)}, \qquad \boldsymbol{\sigma_i^2} = \frac{1}{T}\sum_{\boldsymbol{t=1}}^{T} (p_i^{(t)} _ \boldsymbol{\mu_i})\mathbf{2}$$

The predictive variance σ_i^2 serves as an uncertainty estimate.

Rather than selecting pseudo-labels solely based on inter-expert consistency, the model will additionally require low predictive variance. Only predictions satisfying both:

- High inter-expert agreement
- Low predictive uncertainty

will be used for test-time adaptation.

2. **Uncertainty-Guided Gating Mechanism**

In the original GRAM framework, gating weights determine expert contributions based primarily on learned routing scores (Zhang & Jiang, 2026). The proposed UA-MoE modifies the gating function to incorporate uncertainty-aware weighting:

$$w'_i = \frac{w_i}{1 + \sigma_j}$$

where w_i is the original gating weight and $\sigma_i$is predictive uncertainty. Experts with lower uncertainty receive higher effective weights, improving the reliability of final predictions.

Justification of the Improvement

This enhancement addresses several of the limitations stated in the critical review as follows:

Minimizing Confirmation Bias in Pseudo-Labeling: Pseudo-label refinement assumes that confident predictions are accurate. However, many neural networks are also overconfident. By integrating uncertainty estimation, the model will avoid adapting to predictions based on unreliable assumptions regarding the confidence level of the predictions, which could result in increased error at test time during test-time adaptation.

Enhancing Generalization Across Domains: Domain shifts can lead to an increase in epistemic uncertainty. By integrating uncertainty into gating, the model will be able to decrease reliance on experts who are unsure about unfamiliar regions, which should enhance cross-city robustness.

Ethical Reliability in Social Applications: Slum detection affects vulnerable populations. The ability to provide uncertainty estimates should enhance transparency regarding what was detected and should minimize the likelihood of unjustified high-confidence misclassifications, which is in accordance with responsible AI design principles and enhances deployability in real-world policy contexts.

Practicality and Feasibility of Implementation: Monte Carlo Dropout is a low-cost and computationally efficient method to implement. It can be implemented with little modification to the backbone architecture. Additionally, it can be implemented by simply adding stochastic dropout layers during inference, which makes it feasible for implementation in Part 2, given reasonable computational resources.

Expected Outcomes and Hypothesis

It is proposed that the UA-MoE framework will improve cross-region generalization and prediction reliability. Specifically, the following hypotheses are proposed:

- **Increased Cross-City F1-Score:** By filtering out the pseudo-labels that are unreliable, UA-MoE is anticipated to increase the cross-city F1-scores by approximately 3–5%, relative to the original MoE implementation.
- **Decreased False Positives in New Regions:** The incorporation of uncertainty-aware gating should help to decrease the number of false positives in new regions, especially in those urban layouts that have a combination of different characteristics.
- **Improved Calibration:** It is anticipated that the model will show improved calibration metrics (for example, Expected Calibration Error), which reflects a greater degree of correspondence between the confidence and accuracy of the model's predictions.
- **Reasonable Inference Time Increase:** Although MC Dropout increases the number of inference passes, this should result in only a moderate increase in inference time when compared to the total amount of time required to perform inference for multi-expert architectures.

Part 2 Implementation Plan

For Part 2, the implementation plan will follow this structure:

- Establish baseline CNN performance.
- Implement standard MoE architecture.
- Integrate Monte Carlo Dropout in expert heads.
- Modify pseudo-label filtering using uncertainty thresholds.
- Evaluate performance across cross-city splits using accuracy, F1-score, and calibration metrics.

A structured implementation approach ensures that the implementation process is aligned with the critique, proposed extension, and evaluation process.

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